

The Tor Network and Non-State Actors
An Analysis of the Diffusion of Power in Three Dimensions

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Abstract

The diffusion of power was a phenomenon first described by Susan Strange in the context of International Political Economy (IPE) in 1996. The phenomenon encompasses the concepts of relational and structural power, which are essential for understanding the IPE discipline. Although described in the 1990s, the diffusion of power did not address the cyber domain, nor explicitly the technological revolutions that culminated in the global computer network. In cyberspace, various actors operate to impose their authority over individuals and services. In this context, at the beginning of the 21st century, the low-latency anonymous network known as "The Onion Router" emerged, being part of the so-called "Dark Web" — a region of cyberspace characterized by active efforts to shield communication, privacy, and specific users. This research aims, through journalistic articles published between 2007 and 2017, to analyze the diffusion of power in the TOR network across three dimensions: authority, control, and results.

Keywords: Diffusion of Power. Dark Web. TOR Network. Structural Power. Relational Power. International Political Economy.

Chapter 1

Introduction

The present research falls within an area of study still uncommon within the field of International Relations (IR): one that treats the cyber domain as the new stage for actions among various actors in the International System (IS). This area is part of a broader thematic umbrella that discusses the role of Science and Technology (S&T) within the discipline and seeks to understand their influence on international affairs.

Authors have expressed the importance of S&T for the various dynamics addressed by the discipline of IR (SKOLNIKOFF, 1993; WEISS, 2005; KRIGE; BARTH, 2006; MATTHEWS, 1997). Some emphasize that S&T is a powerful and persistent factor, capable of producing social changes and thus culminating in international affairs (SKOLNIKOFF, 1993; MATTHEWS, 1997)¹. Others point out that this relationship also has an inverse meaning—that is, when S&T themselves are influenced by international affairs, either directly or indirectly (WEISS, 2005, p.297)². That is, just as international affairs are influenced by technological development, the reverse would also be true.

¹Indeed, according to Skolnikoff (1993), after the advent of World War II and the discovery of a myriad of scientific breakthroughs—including the discovery of nuclear fission by the U.S. Manhattan Project—any doubt about the relevance of technology for the purposes of the State was extinguished. In this era, almost all the technology that the world may assimilate or deal with is the result of a calculated production of decisions made within existing political processes. In fact, the author himself questions which aspects, if any, of international affairs have not yet been "touched" by S&T. In particular, Skolnikoff (1993) points to evidence of the evolution of S&T in international affairs: the massive deployment of strategic nuclear forces; the Chernobyl nuclear accident; the monetary transactions occurring abroad through digital financial markets—which, at the time, already surpassed 500 billion U.S. dollars daily; and the near-total eradication of smallpox. Another author, Jessica T. Matthews (1997), in the mid-1990s, pointed out the relative decline of States in the face of the rise of non-state actors on the global stage. These actors were to be "catapulted" onto the global stage due to the telecommunications and computer revolution—the political and social consequences of which, according to her, had until then been virtually ignored by academia.

²Weiss (2005) points out that the direct effects can be felt in four ways: first, public opinion directly affects public support for investments in S&T; second, the goals of foreign policy have the capacity to alter national priorities, reallocating certain investments to the detriment of others; third, the state of relations between two countries affects the migration of scientists, their communications, collaborations, access to study objects, etc.; fourth, international agreements affect the global strength directed toward the protection of intellectual property. Indirect effects operate through mechanisms of Economics, Law, Politics, and Culture.

Despite the relevance of the topic for IR, the academic literature on it remains limited. In this regard, Eugene Skolnikoff stands out as one of the major figures in the convergence between S&T and IR. His work *The Elusive Transformation: Science, Technology and the Evolution of International Politics* (1993) became a reference point regarding the consequences of scientific and technological changes on the evolution of international politics in various dimensions. In this work, Skolnikoff (1993) also reflects on the reasons why the topic of S&T has been neglected by scholars in IR. He suggests, for instance, that information and knowledge arising from S&T are not treated with the importance they deserve because the perception, on the part of IR scholars, is that they are sometimes “obscure” topics. This obscurity is the reason they are relegated to the second or even third plan.³ Author Weiss (2005) complements this view of difficulty in assimilation: according to him, the topic of S&T, when addressed by the field of IR, is approached without a defined logical structure. It is as if the subject were composed, apparently, of residual topics from other themes on the IR agenda.

In the 1990s, the revolution of information systems, as an aspect within S&T, was a relevant point of academic discussion. Regarding authors who reflected on the interface between the theme and IR, four scholars stand out: Strange (1988), Skolnikoff (1993), Builder (1993), and Nye (2011). In common, these authors indirectly pointed out how technological changes can impact State power.

In 1988, before the commercialization of the Internet, the British scholar Susan Strange published *States and Markets*—considered a starting point for understanding structural power and, for this reason, a major milestone in the literature of IR and International Political Economy (IPE). In this work, while discussing the *knowledge structure*, Strange (1988) reflected on the technological revolution emerging at the end of the 20th century. In particular, three considerations stand out: (1) the development of sophisticated computer systems with accessibility (availability and low cost) to the masses; (2) the expansion of communication systems using satellites orbiting the planet; (3) and the aspect of language digitization, which significantly reduces the linguistic barriers separating human groups. In general, Strange (1988) outlined insights into the technological revolution and its implications for the provision and control of information and communication systems. She also highlighted changes in the use of language and non-verbal communication channels; in perceptions she deemed fundamental to the human conditions that influence judgments and, through them, political and economic policies and decisions. These points are essential for understanding the dynamics of the knowledge structure, the diffusion of power, and the impact of these elements on State power.

Another scholar, Eugene Skolnikoff, in 1993—during the threshold period of the com-

³Skolnikoff comments on this when he states that “even scholars concerned with theoretical issues in international relations tend to treat science and technology as static ‘givens’, or as emanating from impenetrable black boxes.” (SKOLNIKOFF, 1993, p.9)

mercial and global opening of the worldwide computer network—did not fail to ponder information technologies. Although he did not explicitly address the Internet and its implications for international politics, he discussed the relevance of recent technological changes to political power. According to him, the introduction of information technologies would considerably increase the limitations on the centralization of political power.⁴ For Skolnikoff (1993), this conclusion is so widely accepted among commentators on international politics that analyses of international affairs, to a large extent, refer to information technologies as key factors that would undoubtedly bring about changes in the international system. (SKOLNIKOFF, 1993). However, he makes it clear that this will not result in the dissolution of the state or a reduction in its relevance in the international system.

At the threshold of the commercial opening of the Internet, Carl H. Builder was another author who contributed to the analysis of the impact of the information systems revolution on international relations. According to Builder (1993), part of the enthusiasm about the new information age stems from the possibilities that the exploitation of hardware offers in relation to power. This era, responsible for shortening the distances across the globe, disseminated a new perspective on power based on information. For Builder (1993), the state faces a dilemma: should it allow or deny free communication through the new information systems? Permitting it would result in granting the power of information to individuals. And, in possession of this power, they would have sufficient capacity to challenge various hierarchies that were themselves established and maintained through the logic of control and denial of information. On the other hand, denying access to communication and the technologies arising from this new information age could mean missing the opportunity to firmly integrate into the global economy.⁵

Finally, in 2011, the renowned IR scholar Joseph Nye Jr. published the work *The Future of Power*, in which he discussed the relevance of the new information age for rethinking power. In this work, Nye reflects on the technological revolutions of the late 20th century and how they gave rise to a new form of power, which he termed “Cyberpower.” It

⁴Especially regarding these limitations, he states that “The effects are easier to see and analyze in those nations, for authoritarian governments, aware of the importance of information to their maintenance of power, have made more conscious attempts to control information flow. But, for all governments, information technologies have similar implications for autonomy, openness, and decentralization of power.” (SKOLNIKOFF, 1993, p. 102).

⁵Builder (1993) also addresses other relevant points on the topic. One of his questions revolves around the apparent “revolution” of information at the end of the 20th century: is the world facing a transition or a revolution? If it were a transition, it would be a shift from a society with relative information poverty to one with relative information abundance. As such, this transition would have revolutionary effects on all existing institutions and hierarchies. However, if the world is facing a revolution, the most obvious victims of this dynamic are nation-states and their governments. This does not mean they would disappear, but they would give way to new actors. Thus, an important consequence of this new information age is the emergence of transnational factions. Similar to Strange (1996), Builder (1993) also discusses the power concentrated in the nation-state and its subsequent diffusion to new non-state actors.

is in this hybrid regime scenario—where the concrete reality of cables, machines, computers, satellites, and other infrastructure merges with the virtual abstraction of digital bytes and interactions among agents—that the fabric of the new domain of the 21st century is woven: the cyberspace. In this realm, the state and new actors emerge to seek space and compete for power. (NYE, 2014). However, he advises caution when considering power in this domain, as cyberspace will not replace geographical space nor extinguish state sovereignty. (NYE, 2010).

In summary, the aforementioned authors, despite making relevant considerations about the ongoing technological changes at the end of the 20th century and their impact on state power, were cautious in asserting that such changes do not signify a reduction in the relevance of the state in the international system (IS), nor its extinction.

Before the commercialization and mass use of the Internet, if there was already discussion about the need to include the topic of Science and Technology (S&T) in the range of issues relevant to the field of International Relations (IR), today this need is paramount.⁶ There is a need for part of the research efforts to focus on the intersection between Information Systems, Science and Technology, and International Relations—considering the development of the topic and its significant public interest, especially in the range of issues focused on cyberspace. The latter, in fact, has instilled political and economic implications significant enough to reverberate in research across different fields of study that intersect with themes on the IR agenda.

When it comes to research and investigations in the field of IR and cyberspace, the topics vary widely. Some research focuses on the technological dimension of the Westphalian system in the Internet age (BRUNN, 1998; MILLS, 2012; TIKK-RINGAS, 2012; DEIBERT, 2013; DEMCHAK; DOMBROWSKI, 2013). Other studies seek to understand IR through the distinctive lens of Information Technologies (IT)—whether analyzing the convergence of the "digital age" and the "age of terror" (DERIAN, 2003); or examining the role of the state, private actors, and online commerce in regulating information flows (FARRELL, 2006). Much has also been discussed about digital governance—with particular emphasis on the investigations by Canabarro (2014). Other researchers advocate for a view of IR within the cyber domain to investigate digital governance (VAISHNAV; CHOUCRI, 2013; MUELLER; SCHMIDT, 2013). Additionally, there are those who discuss the role of the worldwide web in diplomatic and foreign policy issues (ROSS, 2011). After all, since politics is an action linked to the activities of actors in a collective context, much has been pondered about the political activities of these actors in the cyber domain (CASTELLS; CARDOSO, 2005; ROCHE; BLAINE, 2014; SIEDLER, 2016). Above all, perhaps the most investigated topics are two: cybersecurity (NISSENBAUM, 2005; NYE, 2011; CHOUCRI; GOLDSMITH, 2012; KALLBERG; THURASINGHAM, 2012;

⁶Eugene Skolnikoff (1993) is perhaps the leading figure in this agenda that advocates for the inclusion of technology in the issues of the discipline of International Relations.

KRAMER, 2012; HATHAWAY, 2012; CEPIK; CANABARRO; BORNE, 2014) and cyber warfare (BERENGER, 2006; WALSH; BARBARA, 2006; MICHALSKI; GOW, 2007; CLARKE, 2009; LIBICKI, 2009; CORNISH et al., 2010; LAWSON, 2012; HURWITZ, 2013; JUNIO, 2013; CEPIK; CANABARRO; BORNE, 2015; JAJODIA et al., 2015; SCHREIER, 2015; SCHNEIDER, 2016). There is, above all, a significant gap in the study of the Dark Web and IR, this thesis’ topics.⁷

This research is concerned with the structure of knowledge in the context of the cyber domain. Specifically, it seeks to investigate evidence of the diffusion of power through the low-latency anonymous network *The Onion Router* (TOR)—a component of the Dark Web. The latter is a peculiar part of the global computer network, with its own definition. Therefore, we adopt the field of International Political Economy (IPE), from the perspective of Strange (1988, 1996), as the theoretical framework for this study, as we believe this approach has sufficiently relevant elements to explain the phenomenon of the diffusion of power in the cyber domain, encompassing both the plurality of actors and the digital context in which they operate.

1.1 Delimitation of the Theme and Research Question

In her 1996 work, *The Retreat of the State: The Diffusion of Power in the World Economy*, Susan Strange explicitly states that structural power can be observed through an analytical framework that defines “who-gets-what” in global society based on four basic structures: the security structure, the production structure, the financial structure, and the knowledge structure.⁸ In summary, the objective of her work is to draw attention to the power that originates from these structures, particularly from non-state actors. Due to this power, such actors have the capacity to alter outcomes in the International System. Structural power, as defined by Strange (1988), is derived from and conditioned by these four structures.

⁷The SCOPUS database—the largest database of abstracts and citations of peer-reviewed literature, according to its own definition—when the user conducts a search using the keyword “dark web,” restricting the fields to “social sciences,” “decision sciences,” “business, management and accounting,” “economics, econometrics and finance,” “multidisciplinary,” and “arts and humanities,” that is, areas related to IR, the search returns 38 results. When, instead of “dark web,” the keyword “tor network” is used, the result is 24 occurrences. In other words, the total is 62 documents. When restricting the fields to “computer science,” “engineering,” and “mathematics,” using the keyword “dark web,” there are 87 results. The keyword “tor network” provides 165 documents. In total, there are 252 documents—more than four times the results in the social sciences fields.

⁸Within each of these structures, power over others—and also over the mix of values within the system itself—is exercised within and beyond the borders of nation-states by: (1) those who are in a position to threaten or provide security; (2) those who are in a position to refuse or provide credit; (3) those who define what is produced, where, by whom, and under what conditions and terms; (4) those who control access to knowledge and information, as well as those who are in a position to define the nature of knowledge (STRANGE, 1996).

Through this analytical framework on the nature of power, the author managed to distance herself from the traditional currents of the IR discipline that upheld the position of the nation-state as the main actor in the International System—and, according to her, often the only one.⁹ Strange (1996), in turn, seeks to argue for the existence and relevance of other actors beyond the State in global politics and economy: international organizations, transnational corporations, transnational professions, multinational companies, etc.

The focus of our research is the analysis of the diffusion of power in the context of the anonymous TOR network, embedded within the cyber domain. Due to the nature of this network, the diffusion of power is based on the fourth structure defined by Strange—the knowledge structure. Regarding power and this structure, Strange (1988) stated:

[The power derived from the knowledge structure is the one that has been most overlooked and underrated. It [...] comprehends what is believed (and the moral conclusions and principles derived from those beliefs); what is known and perceived as understood; and the channels by which beliefs, ideas and knowledge are communicated—including some people and excluding others. [...] Analysis of the knowledge structure is therefore far less advanced and has far more yawning gaps waiting to be filled, than analysis of other structures. [...] Ordinary people in their everyday wisdom have always recognized that ‘knowledge is power’. But in a rapidly changing global knowledge structure such as we have today it is by no means clear to social scientists who has that power. One trouble is that the power derived from the knowledge structure is often very diffused. (STRANGE, 1988, p.119).]

In the realm of cyberspace, which emerged at the end of the 20th century, it is possible to observe a multiplicity of actors using the Internet to carry out their activities. In this public Internet—free of restrictions and devoid of technologies that provide users and their communications with privacy and anonymity—concerns about surveillance by companies and government entities are voiced by activists and whistleblowers (LANDAU, 2013; SCHEUERMAN, 2014; BAKIR, 2015; WALSH, MILLER, 2015; MURATA, ADAMS, PALMA, 2017). For this reason, the low-latency TOR network—which seeks to provide privacy and anonymity to users and communications through the global Internet—enters the debate on surveillance and privacy in the digital age as an important communication channel. Through the Dark Web, specifically the TOR network, various actors have carried out political operations that have resonated in the media in recent times and have become known to the general public. To illustrate this point, we cite: the WikiLeaks group led by Julian Assange (WIKILEAKS, 2016); the hacktivist group Anonymous (COLLEMAN, 2011); the transnational organization Reporters Without Borders (REPORTERS WITHOUT BORDERS, 2016); former National Security

⁹In particular, Realism, in its various strands, views the nation-state as the holder of power in international relations—whether through the conception of relational power or material power.

Agency (NSA) agent Edward Snowden—who uses the TOR network, among other mechanisms, to safeguard his communications and ensure privacy (LEE, 2015); the terrorist group Al-Qaeda (COHEN-ALMAGOR, 2012); the Silk Road online drug marketplace (CHRISTIN, 2012); cryptocurrencies like Bitcoin, widely used in commercial transactions on the Dark Web's anonymous networks (KUHN, 2015); and, to a lesser extent, jihadist terrorists and pedophiles (MOORE; RID, 2016).

Since, according to Strange (1988), the structure of knowledge comprises the channels through which beliefs, ideas, and knowledge are communicated, it is reasonable to investigate the diffusion of power within the TOR network and the actors who operate power through this channel. Thus, the research question was formulated to understand how, and to what extent, the TOR network increases the diffusion of structural power in the 21st century. The aim of this research is to investigate the issue of the diffusion of power in this region of cyberspace, examining evidence of the increase or decrease of power by non-state agents. Such evidence should, above all, be tied to outcomes in the geographical real-world dimension of the international system (IS).

1.2 Justifications

There are six main justifications that underpin the choice of the theme and scope of this research: first, the rise of the "cyber" theme as a niche for study in International Relations (IR). Second, the urgency of the topic of anonymous networks in the debate on global digital surveillance perpetrated by governmental and corporate entities. Third, the choice of the structure of knowledge over other structures addressed by Strange (1988)—security, production, and finance—is justified. Fourth, the potential academic contributions of this research to the studies on the diffusion of power in the fields of International Political Economy (IPE) and International Relations (IR). Fifth, the choice of the low-latency anonymous network TOR, the selected journalistic media, and the actors presented in this research are also justified. Lastly, the author's personal motivations for the choices and boundaries set in this research.

First, the rise of the "cyber" theme occurred primarily from the new millennium onward. The end of the last century brought about paradigm shifts in technological, political, and social dimensions. According to Castells and Cardoso (2005), this century was responsible for "catapulting" human life into a new phase in technological terms. The period encompassing the significant changes related to the technological "leap" experienced by society is known in the literature as the "Information Age" or the "Network Society." The latter, by definition, "[It is] the social structure that results from the interaction between the new technological paradigm and the overall social organization" (Ibid., p.3). There is a novel element in this society, which is the "microelectronic foundation of network technology that provides new capabilities for an ancient form of social organization:

in networks" (Ibid., p.4). In other words, humans did use the communication networks prior to the 20th century, which were quite common, but the novelty arises from the nature of the network connections, which are now based on digital technology and have a truly global reach (Ibid., p.4).¹⁰

It is necessary to recognize that, at the end of the 20th century, a new domain emerged that was not adequately embraced by the field of International Relations (IR)—despite its global reach and social and economic character. However, even though it does not have the same volume of studies as the traditional areas of the discipline, considerable advances have been made, as previously highlighted in this research. One of the reasons why the "cyber" theme has gained more space within the scope of the social sciences—although still relatively smaller than in the natural sciences—was the publication, in June 2013, by the British newspaper *The Guardian*, of a series of reports that became known as the "Snowden Revelations," the content of which reverberated across various countries simultaneously. The detailed information provided by the newspaper came from secret documents leaked by Edward Snowden, an employee of Booz Allen Hamilton, a company providing services to the National Security Agency (NSA) of the United States of America (USA). According to Landau (2013), the documents revealed numerous surveillance and espionage schemes focused on the Internet.¹¹ From the Snowden Revelations, the academic community published several studies on the subject from different perspectives (Landau, 2013; Chadwick & Collister, 2014; Landau, 2014; Lucas, 2014; Scheuerman, 2014; Toxen, 2014; Bakir, 2015; Branum & Charteris-Black, 2015; Lyon, 2015; Merck, 2015; Nocetti, 2015; Qin, 2015; Salvo & Negro, 2015; Walsh & Miller, 2015; Gürses, Kundnani & Hoboken, 2016; Murata, Adams & Palma, 2017). Authors Castells and Cardoso (2005) state that the Network Society currently constitutes the core of contemporary societies. For this reason, it is imperative that new studies be conducted on the "cyber" theme.

Secondly, issues related to the maintenance of rights and civil liberties that can ensure privacy, consent, and free expression on the Internet gained new dimensions after the massive revelations of digital surveillance perpetrated by the U.S. through the In-

¹⁰Castells and Cardoso (2005) make a small caveat: although considered global, this network society is diffuse and presents an element of exclusion by not including the entirety of humanity.

¹¹Landau (2013) outlines several of them: the domestic telecommunications metadata collection program of Verizon Business Networks Services—including the "who," "what," and "when" of telephone calls; the NSA's "PRISM" program, which targeted specific Internet communications and data storage of individuals outside the borders of the United States and those with whom they communicated; the extent of cooperation between U.S. private companies and the government in providing user and client data; information on U.S. espionage on Chinese computers; the joint operation between the NSA and the Government Communications Headquarters (GCHQ), the British counterpart to the NSA, to monitor the communications of political leaders attending the 2009 G20 Summit in London; the British mass surveillance scheme; the Internet domestic communications metadata collection scheme, among others. Author Lucas (2014) adds more to this: the "XKeyscore" program, revealed in July 2013; and the data chaining program, "Enterprise Knowledge System."

ternet.¹² After the leak of secret documents, Snowden, living in exile in Russia, began advocating for privacy, encryption, and a robust reform of current surveillance systems (Lee, 2015). Privacy and anonymity on the Internet are consistently and openly defended by the former NSA agent. In this regard, the TOR anonymous network has been elevated by him to the status of the most important privacy-enhancing technology of the current era (Ibid., 2015). Snowden’s political stance is not limited to the desire for privacy for ordinary citizens alone. On the contrary, he advocates for other members of society: from those living under authoritarian regimes to professionals committed to reporting and factual accuracy, including political dissidents, whistleblowers, activists, and more. Although the actions of these individuals take place in the abstract realm of cyberspace, their consequences have implications for concrete reality. This demonstrates the political nature of the TOR communication channel in the 21st century. Understanding its functioning, as well as the actions of the various agents operating within it, allows an examination of how power relations are configured in the digital realm today—and, once again, with implications for concrete reality.

Thirdly, our approach aims to analyze the diffusion of power within the context of the TOR anonymous network. This network, by definition, falls under the structure of knowledge, as it is this structure that deals with “the channels by which beliefs, ideas and knowledge are communicated—including some people and excluding others” (Strange, 1988, p. 119). For this reason, and for the purpose of defining the scope of our research, we will analyze the diffusion of power through this structure. It is therefore important to understand both the focus of our analysis and the origins of structural power—two distinct points. While the focus of this research’s analysis of diffusion of power rests on the structure of knowledge—which we believe is best suited to explain the reality of the TOR network and the actors operating within it—the structural power, as conceived by Strange (1988), remains a product derived from the four primary structures, to varying degrees. Thus, we believe it is possible for the focus of analysis to be on one of the structures without compromising conceptual meanings. Strange herself, at times, focuses her analysis on different structures when addressing the diffusion of power.¹³

Fourthly, we believe that this research contributes to academic studies on the theme of diffusion of the power in two ways: first, by bringing the theme into the cyber context; and second, by operationalizing the concept in a way that aids analyses of diffusion of

¹²The nonprofit organization Electronic Frontier Foundation (EFF), founded in 1990, is a leading advocate for civil liberties in the digital world and is currently battling the NSA in U.S. courts over the agency’s surveillance of Internet infrastructure (Tummarello, 2016). In court, the EFF argues that the NSA violates the Fourth Amendment of the U.S. Constitution by copying and searching data collected from the Internet backbone within the United States (Maas, 2014).

¹³For example, Strange (1996) focuses on the Italian Mafia within the security structure. The Mafia competes with the Italian state for positions within the security structure. Similarly, she acknowledges that the Mafia’s structural power is derived, to a greater or lesser extent, from all four structures—whether through providing physical protection to members and allies, trading illicit goods and services, offering credit to interested parties, or accessing knowledge and information networks.

power. Strange’s works (1988, 1996) did not provide variables that could at least guide the direction of analyses on diffusion of power—how it occurs and to what extent. We believe that constructing variables that can at least delineate the analysis is relevant. Therefore, we will work with three variables during our investigation of diffusion of power within the context of the TOR low-latency anonymous network: authority, control, and outcomes. The methodology involving these variables will be described in the methodological section of this introduction.

Fifthly, we justify the investigation of the TOR network, the selected journalistic media, and the actors presented in this research. Within the context of the networks that make up the Dark Web, the TOR network is considered the largest in terms of nodes and users, as well as being perceived as having the most sophisticated cryptographic technology (Moore & Rid, 2016, p. 15). Moreover, we aim to use alternative journalistic media in the form of online newspapers to build the database on which we apply the variables described above. The selection of these newspapers was based on the *Top 10 Online Newspapers Worldwide Ranked by Unique Visitors*, published in 2012 by ComScore, a U.S.-based company headquartered in Virginia that conducts various analyses related to the Internet domain (ComScore, 2012). Finally, the agents selected for the analysis of the three variables were those who: (1) could be considered part of the knowledge structure, according to Strange’s (1988) definition; and (2) actors whose operations took place through the TOR anonymous network.

The sixth and final justification relates to the author’s personal motivations. These motivations can be summarized as an affinity for the themes of computing, information systems, and technology in general. During childhood, the author learned programming languages and, for the first time, was able to build a website on the World Wide Web (WWW). This experience led her to pursue studies in Computer Science, which were not completed due to her enrollment in Civil Engineering. Her interest and affinity for the Exact Sciences have been long-standing. Both fields were left behind when she began and completed her undergraduate studies in International Relations at the Federal University of Santa Catarina, earning her degree in February 2015. Her final undergraduate project (TCC), though brief, focused on the “Deep Web” and the “Dark Web.” The TCC was supported by three justifications: the latent interest in Technology and Information Systems, especially the world of the Internet; the activist movements in the cyber domain such as Anonymous, WikiLeaks, the TOR Project, and Edward Snowden’s revelations about global communication surveillance carried out by the Five Eyes members, which had political implications for the international scenario; and, finally, the growing concerns regarding the marginalization of these topics within the field of International Relations (IR). In 2016, she enrolled in the Graduate Program in International Relations at the Federal University of Santa Catarina with the intention of continuing her studies on the Diffusion of Power and the Dark Web. In 2017, she published the book *Deep&Dark Web:*

The Internet You Know Is Just the Tip of the Iceberg through Alta Books company.

This research is, above all, an attempt to analyze the forces and movements that occur in the cyber domain and reverberate in international political economy and in the daily lives of millions of people around the world.

1.3 Objectives

This dissertation works with one general objective and five specific objectives.

1.3.1 General

The general objective of the research is to understand how, and to what extent, the TOR network increases the diffusion of structural power in the 21st century. To enable the achievement of this general objective, the following specific objectives are proposed.

1.3.2 Specific

1. To examine the theoretical aspect of structural power that fundamentally diverges from traditional approaches to power in International Relations (IR);
2. To propose the operationalization of the concept of diffusion of power, developed by Strange (1996), through three ordinal qualitative variables;
3. To describe the historical, theoretical, and technological aspects that support and accommodate the anonymous TOR network within the cyberspace universe and distinguish it from other communication channels;
4. To analyze the database—composed of subjects from the structure of knowledge selected through newspaper articles—in order to assign values to the variables representing the three dimensions of power that confirm its diffusion;
5. To evaluate, through the previously described specific objectives, how and to what extent the TOR network increases the diffusion of power in the 21st century.

1.4 Methodology

The methodological starting point of this dissertation is the literature review. Initially, we start with a holistic view of the theme of "Science", "Technology", and "International Relations" and the positioning of this research within the aforementioned theme. Subsequently, we map the discussion on the Internet and International Relations in order to investigate the issues already addressed by scholars in the field regarding the cyber

domain and correlate them with this research (as presented in the introduction of this dissertation). We conducted an extensive search on "Power" and "International Relations" to examine how the theoretical framework related to power is configured as an approach within the discipline. Next, we observed how the element of "Diffusion of Power" is addressed by the theoretical framework and other authors in the field—aiming to outline distinctions and similarities. The theme of this research, Dark Web, is situated in cyberspace, and for this reason, we sought to identify the worldwide computer network as its own domain. To achieve this, we explored the literature on the "Origin of the Internet" in its technical aspect to gather knowledge about its operationalization and functioning. Consequently, we understood from the literature on the Deep Web the different nuances of the World Wide Web, among which the Dark Web is configured as just one of the categories. Finally, for the purpose of understanding the specificity and technological nature of the low-latency anonymous TOR network, we examined the literature on the Dark Web and TOR in order to provide the basic inputs through which we conducted the analysis of diffusion of power.

Below is a table composed of the literature review.

Table 1.1: Literature Review

Categories	Literature on the subject
Science, Technology and International Relations	SKOLNIKOFF, 1993; WEISS, 2005; KRIGE; BARTH, 2006; MATTHEWS, 1997; BUILDER, 1993.

Categories	Literature on the subject
Internet and International Relations	NYE, 2011; LANDAU, 2013; SCHEUERMAN, 2014; BAKIR, 2015; WALSH, MILLER, 2015; MURATA, ADAMS, PALMA, 2017; BRUNN, 1998; DERIAN, 2003; CASTELLS; CARDOSO, 2005; NISSENBAUM, 2005; BERENGER, 2006; WALSH; BARBARA, 2006; FARRELL, 2006; MICHALSKI; GOW, 2007; CLARKE; 2009; LIBICKI, 2009; CORNISH et al, 2010; ROSS, 2011; LAWSON, 2012; CHOUCRI; GOLDSMITH, 2012; KALLBERG; THURAISINGHAM, 2012; KRAMER, 2012; HATHAWAY, 2012; MILLS, 2012; TIKK-RINGAS, 2012; HURWITZ, 2013; JUNIO, 2013; DEIBERT, 2013; DEMCHAK; DOMBROWSKI, 2013; VAISHNAV; CHOUCRI, 2013; MUELLER; SCHMIDT, 2013; ROCHE; BLAINE, 2014; CANABARRO, 2014; CEPIK; CANABARRO; BORNE, 2014; CEPIK; CANABARRO; BORNE, 2015; JAJODIA et al, 2015; SCHREIER, 2015; SCHNEIDER, 2016; SIEDLER, 2016; CHADWICK, COLLISTER, 2014; LANDAU, 2014; LUCAS, 2014; SCHEUERMAN, 2014; TOXEN, 2014; BAKIR, 2015; BRANUM, CHARTERIS-BLACK, 2015; LYON, 2015; MERCK, 2015; NOCETTI, 2015; QIN, 2015; SALVO, NEGRO, 2015; GURSES, KUNDNANI, HOBOKEN, 2016; MURATA, ADAMS, PALMA, 2017; TUMARELLO, 2016; CASTELLS, 2001.
Power and International Relations	PORTER, 2013; WRIGHT, 1995; DE JOUVENEL, 1957; BALDWIN, 2013; HENDEL, 1953; WALTZ, 1979; WALT, 1987; MOUL, 1989; CLAUDE, 1989; GUZZINI, 2000; SCHWELLER, 2006; KAUFMANN, LITTLE, WOHLFORTH, 2007; LITTLE, 2007; BROOKS, WOHLFORTH, 2008; WENDT, 1999; MORGENTHAU, 1948; SHEHAN, 1996; POLLARD, 1923; PAUL, 2004; WALTZ, 1990; MEARSHEIMER, 2001; SIMONDS, EMENY, 1937; LASSWELL, KAPLAN, 1950; NAGEL, 1975; DAHL, 1961; BACHRACH, BARATZ, 1962; LUKES, 1974; WENDT, 1992; HOPF, 1998; BARNETT, DUVAL, 2005; GUZZINI, 2007; KEOHANE, NYE, 2011; BALDWIN, 1995;
Diffusion of Power	BUILDER, 1993; NYE, 2011.

Categories	Literature on the subject
Internet Origin	FINKLEA, 2015; CASTELLS, 2001; NAUGHTON, 1999; BARAN, 1964; LAKSHMAN, MADHOW, 1997; CERUZZI, 2003; ASPRAY, CERUZZI, 2008; BANKS, 2008.
Deep Web	BERGMAN, 2001; FIDÊNCIO, MONTEIRO, 2013; SHERMAN, PRICE, 2001; BECKET, 2009.
Dark Web and TOR	SHERMAN, PRICE, 2001; FIDÊNCIO, MONTEIRO, 2013; BECKET, 2009; GEHL, 2016; CHERTOFF, SIMON, 2015; DEVINE, EGGER-SIDER, ROJAS, 2015; FINKLEA, 2015; ROCHE, 2016; ZULKARNINE et al, 2016; GHAPPOUR, 2017; MOORE, RID, 2016; DINGLEDINE, MATTHEWSON, 2005; PACHENKO, PIMENIDIS, RENNER, 2008; EDMAN, SYVERSON, 2009; CHAABANE, MANILS, KAAFAR, 2010; HOPPER, VASSERMAN, CHAN-TIN, 2010; LI et al, 2010; LI et al, 2011; ALSABAH, BAUER, GOLDBERG, 2012; MOGHADDAM et al, 2012; EDMAN, SYVERSON, 2009; LOESING, MURDOCH, DINGLEDINE, 2010; ELAHI et al, 2012; MCCOY et al, 2008; DINGLEDINE, MATTHEWSON, SYVERSON, 2007; AKHOONDI, YU, MADHYASTHA, 2012; MITTAL et al, 2011; MOGHADDAM et al, 2012; ALSABAH, BAUER, GOLDBERG, 2012; DUNGHEL et al, 2010; JOHNSON et al, 2013; FIFIELD et al, 2012; EDMAN, SYVERSON, 2008; HORSMAN, 2017; THOMSON, 2017; SCHULZE, 2017; BUCHANAN, 2017; SANDVIK, 2017; BIDDLE et al, 2011.

The theoretical framework of this dissertation is based on the works of the British scholar Susan Strange, which directly and indirectly address the phenomenon of the diffusion of power: *States and Markets* (1988) and *The Retreat of the State* (1996). From this theoretical framework, the need arose to systematize its content in the form of three distinct and complementary dimensions that make up the phenomenon of the diffusion of power: authority, control, and outcomes. These dimensions are implicitly present in Strange's works—but not as formal dimensions per se; rather, they serve as relevant elements for the construction of the analytical framework through which the author's global political economy vision is solidified. The diffusion of power is a phenomenon that occurs separately in each of these three dimensions.

The element of "authority" is interpreted through the lens of the concept of power. On rare occasions, Strange (1996) uses the term "power" when discussing the phenomenon of the diffusion of power, preferring instead the term "authority." When examining the

term 'authority' in her work, we notice a duality of interpretation attached to it: on some occasions, the term is seen as the institution, entity, or subject that embodies the expectation of power (being authority); on others, it is viewed as synonymous with power, or more precisely, with the ability to affect outcomes in such a way that the subject's preferences are imposed (exercising authority). In the first chapter, we aim to substantiate the reasons that indicate the existence of this duality and provide more detailed considerations about how these "authorities" manifest.

In this dimension, Strange discusses the effects of power arising from the presence or involvement of an authority—without the need for the authority to act, operate, or perform activities. Strange (1996) argues that this conception of power finds its origins in feminist approaches. Within a specific context, the presence of a figure understood as authority can be representative of power—and produces effects that are felt and perceived by other figures within that context, leading to consequences. To illustrate, the author addresses the relations between men and women and the inherent power of the male presence in the social context.¹⁴ Furthermore, the presence of authority can emanate from its peers, recognizing three possible scenarios: that the authority is perceived as stronger, has remained the same, or has weakened.¹⁵ We will discuss these ideas in detail

¹⁴Especially regarding this notion that power can act merely through presence—and not solely through actions—Strange offers some considerations about this type of power, which she discusses as being "structural", or rather, as a product of the dynamics of established structures (this includes the feminist approach, because power originates from the structural dynamics established in the social context of the patriarchy). She affirms: "A second general point is that 'power over' need not be confined to outcomes consciously or deliberately sought for. Power can be effectively exercised by 'being there', without intending the creation or exploitation of privilege or the transfer of costs or risks from oneself to others, for instance. This recognition of unconscious power is one contribution that gender studies has surely made to international political economy. Male partners may not wish or intend the control they have over outcomes affecting their female partners. But as many women are acutely aware, the social structures within which the partnership exists will make sure that such power exists. [...] This is where the distinction between what I have called relational power and structural power is relevant. In relations with others, it is much harder to think of power being exercised by one party over another unconsciously, without deliberate intent. But when you think of power in terms of power over structures, it is easier to understand that relations existing within those structures are affected, even though it may be inadvertently. The same is true of the power of United States government agencies over outcomes in the international system." (STRANGE, 1996, p.26)

¹⁵The perception of peers regarding authority—whether it is increasing, decreasing, or indifferent—is an idea indirectly addressed in Strange's 1988 work. The author provides several examples and I quote: "Without the productive power to supply food and capital goods for the reconstruction of European industry, and without the financial power to offer credits in universally acceptable dollars, the United States could not have exercised the power over the recipients of Marshall Aid that it did. Nor was American structural power based only on dominance of the security structure, the production structure and the financial structure. Its authority was reinforced by the belief outside America that the United States fully intended to use its power to create a better post-war world for others as well as for its own people. [...] Moral authority based on faith in American intentions powerfully reinforced its other sources of structural power. [...] A very different example of the power derived in part from the force of ideas would be that exercised within and beyond Iran after the fall of the Shah by Ayatollah Khomeini and his followers. The idea that the Shah, out of greed and lust for power, had fallen captive not only to a foreign country but to a culture and a materialistic belief system alien and inimical to traditional Islamic values had contributed powerfully to the collapse of his government and his own exile. But the power of the ayatollahs in defending and promoting Islamic virtues would have been constrained if they had not

in Chapter One, which focuses on the dimension of authority.

The "control" dimension appears frequently in Strange's works (1988, 1996) when discussing the operationalization of objects within primary structures. She often states that actors positioned in primary structures have power when they control aspects relevant to influencing outcomes in the economic-political scenario (Ibid., 1988). However, the controller of the object influencing outcomes is not necessarily the authority. In the first chapter, we identify various examples from Strange's work where actors within primary structures control objects deemed relevant for influencing outcomes.

The "outcomes" dimension is fundamental in Strange's literature because it determines the authority and control of actors: altering the status quo indicates power, while its preservation may indicate a lack of power.¹⁶ This dimension encompasses both structural power (indirect) and relational power (direct).

In sum, Strange's works indicate that the diffusion of power operates through the three dimensions of "authority", "control", and "outcomes". These dimensions culminated in three ordinal quantitative variables present in our database, compiled from journalistic articles, which underpin our analysis of diffusion of power.

Our methodology is structured as follows: we will outline the three variables used in the diffusion of power analysis, explain their operationalization and scope, discuss the selection of journalistic articles for our database, identify the actors derived from these articles, and detail the process leading to our analysis of diffusion of power through three dimensions.

The independent variable is the TOR Network and the dependent variable is "diffusion of power". Our hypothesis is that the TOR Network enhances diffusion of power in the cyber domain. This research focuses on "how" this diffusion is possible and "to what extent" it occurs.

also gained control over the state and the armed forces sufficient to confirm their authority both within the country and beyond. Undoubtedly, the power of ideas was indispensable but it could only be used to affect outcomes in conjunction with military capability and economic resources." (STRANGE, 1988, p.32) I argue this excerpt highlights how moral and ideological authority can reinforce structural power, and how peer perception plays a crucial role in legitimizing and maintaining that authority.

¹⁶We argue that the absence of change does not always indicate an absence of power. Denying access to information, for example, demonstrates power by the actor who withholds it. Strange (1988) highlighted this concept as "negarchy", referring to power derived from resisting changes to the status quo. A classic example is the system of "checks and balances" that operates within the legislative, executive, and judicial branches of government. The ability of one branch to prevent excessive influence from another is indicative of its power. Another example from this research is the Tor Project, which exercises power by resisting digital surveillance over users and messages within the anonymous TOR network. This exercise of power affects outcomes by preventing surveillance, but it does not alter the status quo, as the status quo is the absence of surveillance. A change in the status quo would mean a new reality where digital surveillance is effectively conducted.

1.4.1 Variables "authority", "control" and "outcomes"

It is known that when discussing the diffusion of power, Strange (1996) did not use variables that could substantiate the phenomenon being studied. Her methodology, identified by her as "functionalist", examines the various functions of authority in political economy and raises questions about "who", or "what", would be exercising these functions or responsibilities—and, furthermore, with what effects on the outcomes (STRANGE, 1996, p.42). Although she did not approach the phenomenon from the perspective of operationalization, we believe that constructing ordinal quantitative variables is an important attempt to aid in the interpretation of the phenomenon. These variables are able to, at the very least, guide our analytical approach in relation to the diffusion of power.

Based on the reading of her work, our understanding of the diffusion of power is as follows: diffusion of power occurs when an actor has sufficient authority within a primary structure, at either the local or global level, to control an element that is capable of affecting the outcomes. In other words, their control over this element is decisive in determining the outcomes.¹⁷ According to Strange (1996), the diffusion of power helps explain the rise of market power over the nation-state—or more precisely, the rise of the power of non-state actors who operate within each of the primary structures, and sometimes at the intersection of them (secondary structures). Her approach underscores the relevance and strength of non-state actors in the international system and highlights the urgency of addressing them in academic research to better understand international practice. For this reason, she distances herself from the discipline of International Relations—which is almost exclusively concerned with the state actor—and inaugurates International Political Economy, which encompasses non-state actors operating transnationally within the global political economy.

The table below systematically presents the three variables adopted in this research (first column), their scope of action (second column), and plausible values (third column).

Table 1.2: The Diffusion of Power's Variables

Variables	Scope of Action	Ratings		
Authority	Trust of peers and stakeholders	Decrease (-1)	Stability (0)	Increase (+1)
Control	Exercise/operation of the element	None (0)	Partial (+1)	Absolute (+2)
Outcomes	Impact on the status-quo	No change (0)	Change (+1)	-

Source: Author.

The first column, "variables," identifies the three variables of the diffusion of power

¹⁷For instance, she points out that the State holds control over the Armed Forces, which at times act in a way that influences the outcomes so that the preferences of the State take precedence over others. The control over the Armed Forces (and other structures) underpins the State's supreme authority within the security structure. And although the State faces rivals (for example, organized crime—sometimes referred to as "the parallel state") within the security structure, it still remains within this structure as the leading authority. (STRANGE, 1988).

phenomenon: authority, control, and outcomes.¹⁸ The second column, “scope of action,” presents the meaning or the underlying idea behind each variable, i.e., the scope of action of the variable. The third column, “ratings”, represents through maximum and minimum values all the possibilities encompassed by each variable. For example: the “authority” variable has as its scope the trust of peers and stakeholders. This trust can decrease, remain stable, or increase. If a decrease is observed, based on the analysis of the journalistic article, the “authority” variable will be assigned the value “-1”. And so on. The ratings for each dimension are based on the works of Susan Strange. Through them, it was noted that in situations where she mentions authority, she often analyzes whether a particular authority is in decline, growing, or stable. At other times, Strange (1988) indicates the importance of controlling resources that can impact results. And on several occasions, she emphasizes the importance of outcomes in determining the authority, control, and power of an actor. The foundation supporting the existence of the variables, scope of action, and ratings presented in this table is explained in Chapter One of this research and is backed by the literature of Susan Strange.

The first variable, “authority,” refers to the sense of trust conferred by peers and stakeholders.¹⁹ We can say that something or someone possesses sufficient authority within a structure when their peers grant them trust, whether formally or informally. In other words, before anything else, we need to identify the agent/actor in question as a local or global authority within at least one of the four primary structures identified by Strange (1988). If this is possible, the explanatory sequence of the variable follows. Otherwise, we cannot associate the actor with the “authority” variable. Let us consider an example.

A dentist has authority over dental matters because their professional council has granted them this authority through membership. They also have authority over these matters because the university from which they graduated conferred upon them a bachelor’s degree, which allows them to enjoy certain legal prerogatives—including this authority. A drug trafficker has authority over drug transactions in a region because their peers, other drug traffickers, or stakeholders, such as consumers of the product, have granted them this trust. While the dentist is a formal authority, the drug trafficker is an informal authority. This “authority” can increase (in the case of a dentist, this happens when they conduct in-depth research on topics in their field, becoming a specialist in a specific subject); decrease (when the same dentist commits serious errors in the practice of their profession, leading to fewer clients as trust in their work diminishes; if the severity of the error is confirmed, they may be expelled from the professional council and have their legal prerogatives suspended); or remain stable (when the dentist does not become

¹⁸These variables are representative of the dimensions of the diffusion of power. In turn, these dimensions originate from the works of Strange (1988, 1996), which serve as the theoretical framework for this thesis.

¹⁹Strange uses the term “reliability”.

a specialist in any specific subject but also does not commit errors in their professional practice—that is, they simply maintain their level of authority in the exercise of their professional activities).

An actor has power when they possess authority over something. This authority may have been granted through formal or informal means. Let us examine two examples to illustrate this. To demonstrate formality, we will use the example of sovereignty in the International System (IS). Sovereignty is a power exercised in the IS because it is formally recognized by the actors that compose it, or rather, by the peers of the state (state actors) and stakeholders (non-state actors). Regarding informality, consider the example of a drug trafficker in a region. The local population grants them authority based on community belief, either because they provide security in exchange for the ability to market their products or because they are excessively violent in the region, and the community submits to them out of fear of retaliation. Similarly, the US dollar is a currency with power that extends beyond the United States' borders because other states informally grant it authority when they use this currency in their commercial and financial transactions.

When the “authority” variable increases, we assign it a value of “+1”; when it decreases, we assign it a value of “-1”; and when it remains stable, we assign it a value of “0,” as there is no change. This variable, therefore, can take integer values ranging from “-1” to “+1” (including the value “zero”)—depending solely on the context of the situation in which the authority in question is found and the analysis derived from this context.

The second variable, “control,” refers to the sense of exercising or operating a specific object relevant to achieving influence over outcomes. Again, we must first identify whether the agent has control over the object in question. If this can be ascertained, the explanatory sequence of the variable follows. Otherwise, we cannot attribute the “control” variable to the agent.

An actor has power when they have control over something relevant to achieving their desired outcomes. This power is the ability to alter the status quo. They may not be a formal or informal authority; however, if they have the capacity to control this relevant object, then they possess power over the outcomes. This would be the case, for example, of a military pilot in command of a combat aircraft. They do not possess authority; they are an instrument of operation for the Armed Forces to which they are subordinate. However, when, while flying over a region, they are ordered to release the bombs stored in the aircraft, this pilot has the choice to launch the bombs or not. At that moment, they have control over the object relevant to influencing the outcomes: the bombs. During this period, this soldier has power. Their actions can alter the status quo (e.g., not launching the bomb) if they so choose.

Something, or someone, only controls an object if they perform or operate actions to

that end—whether directly (themselves) or indirectly (using another person or mechanism). The “control” variable, when present, determines the degree of operation over the object: none, partial, or absolute. When this variable cannot be attributed to the agent in question, we say that the agent has no control over the specific object and assign the value of “none,” which corresponds to “0,” as their control is nonexistent. If the agent has partial control, they will receive the value of “+1.” And if they have absolute control, they will be assigned the value of “+2.” This variable, therefore, can take integer values ranging from “0” to “+2”—depending solely on the degree of control exercised by the agent in question.

The third and final variable, “outcomes,” concerns the sense of “impact on the status quo.” And, when it comes to the cyber domain, it is relevant to consider the “effects on reality”—the domain in which the actors are situated. Relevant outcomes are those that impact the real geographic plane—not the virtuality of the cyber domain. After all, an outcome only has meaning if it affects some aspect of the reality in which the actors are embedded. Any outcome in the cyber plane will only be taken into account if it also impacts the real geographic plane. A virtual game for leisure can have different outcomes for the players involved. For example, Player A might finish in first place. However, this outcome has no impact on the real geographic plane, as it is played for pleasure. If, by chance, the game took place within the context of a national championship, the first-place finish would certainly have some impact on Player A’s real plane: monetary compensation, a prize, or a title.

An actor can also have power when their actions, with effects on the real geographic plane where the other actors are situated, alter or reinforce the status quo.²⁰ This is the case of a player in virtual games within a championship. When they compete against the world champion, they have neither authority nor control over the game. However, when they change the outcome (by defeating the world champion, for example), they acquire power (now, they themselves are the world champion). Not altering the outcome (the world champion remains the world champion)—that is, reinforcing the status quo—is also a way of allowing their opponent to maintain the power they already possessed.

The “outcomes” variable can indicate a change in the status quo or its lack of change. If it indicates a change, we assign the value “+1” because the status quo has been altered. If it does not indicate a change, we assign the value “0” because the status quo remains unchanged. This variable, therefore, can take integer values ranging from “0” to “+1” (including “zero,” evidently)—depending solely on the analysis of the change in the status quo in question.

It is necessary to point out that our understanding of the operationalization of the

²⁰As mentioned earlier, the power derived from reinforcing the status quo is related to the idea of “negarchy,” or the force that arises from the denial of changes to the status quo. Strange (1996) reflects on “negarchy” in her work.

phenomenon of the diffusion of power does not concern the gain, stability, or loss of **absolute power** but rather the **level of power**. These are different things. When an actor experiences a loss in their level of power, they lose some power. They do not necessarily lose all their power. But to lose some power, they must have some power to begin with. This is why the situation of having no power and, furthermore, losing a level of power cannot occur. There is no negative power. Now, if an actor experiences stability in their level of power, this means there is neither gain nor loss of power. The actor simply maintains the same level of power they previously held. If they had no power, they remain with none. If they had some power, they retain the same “amount” of power. Their level, wherever it may be, remains unchanged. Finally, if there is a gain in the level of power, this indicates that the actor has necessarily gained some power. If they had none, they now have some. If they already had some, they now have even more—their power is reinforced and reaffirmed.

Finally, we need to make two observations regarding the operationalization of the concept adopted in this research. First, we do not propose that this operationalization, along with the variables, serves as a definitive guide to the diffusion of power. However, we believe it can support a perceptive analysis of the gain, loss, or stability of power in the International System (IS), according to the approach of International Political Economy (IPE). The central idea of this operationalization is to track the journalistic articles selected in this research, in which the actors’ actions are carried out via the anonymous TOR network. We hope this operationalization aids in the analysis of each agent: whether they are losing, gaining, or maintaining the same level of power. Each case can only be evaluated individually, and its context must be respected. The assignment of values to the variables allows, for example, statistical correlation analyses, which, although not used in this research, provide inputs for various analyses. We are also aware that the operationalization of the phenomenon should not be a rigid guide on the subject. Second, the assignment of values to each actor will be based on perception—a subjective characteristic. Each actor, derived from the journalistic articles, will be analyzed through the variables discussed here to frame the perception of the diffusion or power in the context of the TOR network. The actors, who comprise the database of this dissertation, will be identified and discussed in the next section.

1.4.2 Selection of News

The search for analytical inputs has an investigative documentary nature, as it was conducted using documents preserved within private journalistic institutions. According to Gill (1991), this type of research has advantages as it is considered a “rich and stable source of data”, does not imply high costs, and does not require contact with research subjects. Pádua (1997, p.62) further argues that documentary research is conducted

using contemporary or retrospective documents, considered scientifically authentic (not fraudulent), and has been frequently used in the social sciences.

We utilized alternative media—more precisely, online newspapers—to compose the reality framework on which the analysis of the diffusion of power was conducted. Riffe, Lacy, and Fico (2008) discuss media research and point out that the scientific method corresponding to "the systematic assignment of communication content to categories according to rules, and the analysis of relationships involving those categories using statistical methods" is called quantitative content analysis. This research method involves a series of techniques, including theoretical literature examination and the development of representative coding categories—aimed at reflecting differences between contents. Its application to mass communication content, such as newspapers, is based on the initial premise that the researcher seeks to develop a social science-based approach through empirical observations and measurements. In other words, there is sufficient suspicion to investigate "theoretical traits" in a specific practical context and thus propose explanations or relationships between different concepts (RIFFE; LACY; FICO, 2008). Regarding this, Riffe, Lacy, and Fico (2008) provide an example:

If members of an ethnic minority, for example, voice concern that they are underrepresented in news media content (in terms of their census numbers), a researcher may propose that racism is at work or that members of the ethnic minority are underrepresented in those occupational groups that serve more often as news sources in the news. Each of these interpretations or explanations involves different concepts that can be “operationalized” into measurement procedures, and each can be tested empirically, as researchers did in the content analyses highlighted previously. (RIFFE; LACY; FICO, 2008)

The present research involves the operationalization of concepts and empirical analysis through the documentary investigation of journalistic articles and categorical measurements—therefore, it employs the scientific method of quantitative content analysis as presented by the aforementioned authors.

Our investigation was limited to the ten most-visited English-language online newspapers, as presented in the justification section. (COMSCORE, 2012).

The British newspaper *Mail Online* is the most accessed by users worldwide, closely followed by the American *The New York Times* and, again, a British newspaper, *The Guardian*. Two Chinese newspapers (*People’s Daily Online* and *Xinhua News Agency*) and one Indian newspaper (*Tribune Newspaper*) are among the ten largest newspapers by user clicks worldwide—although the list is dominated by Western newspapers²¹. According to the ComScore (2012) report, *Advance Digital* is listed in the last position. Despite

²¹It was not possible to obtain any results for the term “Tor Network” in the newspaper *Xinhua News*

Table 1.3: **The Ten Largest Newspapers in the World by Clicks per User (in Millions)**

Rank	Country	Newspaper	Unique Visitors/Month (millions)
1st	UK	Mail Online	50.067
2nd	USA	The NY Times	48.695
3rd	UK	The Guardian	38.931
4th	India	Tribune Newspapers	35.862
5th	China	People's Daily Online	33.026
6th	UK	Telegraph Media Group	30.083
7th	China	Xinhua News Agency*	29.987
8th	USA	Washington Post	26.007
9th	USA	Hearst Newspapers	24.174
10th - Advance Digital (Group of 13 Newspapers) - 22.340 million visitors			
	USA	Alabama Local News	http://www.al.com
	USA	Cleveland OH	http://www.cleveland.com
	USA	Mississippi and Gulf Coast	http://www.gulfive.com
	USA	Le High Valley	http://www.lehighvalleylive.com
	USA	Mardi Gras	http://www.mardigras.com
	USA	Massachusetts Local News	http://www.masslive.com
	USA	Michigan Local News	http://www.mlive.com
	USA	New Jersey Local News	http://www.nj.com
	USA	New Orleans, LA Local News	http://www.nola.com
	USA	Oregon Local News	http://www.oregonlive.com
	USA	Central Pennsylvania Local News	http://www.pennlive.com
	USA	Staten Island NY Local News	http://www.silive.com
	USA	Syracuse NY Local News	http://www.syracuse.com

Source: comScore MMX, Worldwide, Age 15+, Oct 2012.
*Chinese News Agency.

being classified as a newspaper, it comprises a conglomerate of thirteen American newspapers. In total, the ranking of the ten largest newspapers in the world by user clicks, compiled by ComScore (2012), consists of 22 newspapers with active URLs.

To build the database for this research, we accessed the “archives” section (a publicly available database) of each of the listed newspapers and searched for the expression “tor network” (in quotation marks). With the exception of China’s official news agency (*Xinhua News Agency*), all other newspapers returned a list of search results.²² In total, 302 newspaper articles appeared as results for the search term “tor network.” Of these,

Agency, the official news agency of the Chinese government. The search returned some page display issues, appearing to enter a “loop”—when the user makes a request to access the server unsuccessfully, and the procedure repeats indefinitely.

²²Instead of returning results for the search of the cited term, the newspaper "Xinhua News Agency" repeatedly attempted to access the server but failed to display the results page. For this reason, it was not possible to identify the number of results for the search term "tor network" in the news archives.

only 195 actually dealt with the TOR anonymous network—190 were original articles, while 5 were duplicates.²³ In summary, out of the initial 302 newspaper articles, only 125 simultaneously addressed topics related to the knowledge structure.

Thus, we sought non-state actors who had some degree of social recognition regarding the possession of information and/or knowledge (Strange does not differentiate between these two terms by definition) that were responsible, in some way, for storing information, or controlled the channels through which this information was communicated. Within the universe of 190 original articles related to the TOR anonymous network from the ten largest newspapers by user clicks worldwide, we found that 125 articles discussed non-state actors who met the aforementioned definitions. In some articles, more than one non-state actor was mentioned—so we opted to call “occurrences” the situations involving a single non-state actor and “article” the journalistic reports that could discuss one or more actors. In 14 articles, we identified two actors. Each actor was treated as a unique occurrence deserving analysis.²⁴ In total, we identified 139 occurrences eligible for analysis—these became the 139 rows in the database for this research. In other words, each row in the database represents a single occurrence involving only one non-state actor positioned within the knowledge structure. Through the newspaper articles, we observed that these actors carried out specific actions via the TOR anonymous network.

Table 1.4: **Number of Occurrences for Analysis (2007-2017)**

Occurrences*	N
Results for "Tor Network"	302
Content Unrelated to "Tor Network"	105
Content Related to "Tor Network"	195
Originals	190
Actions via Other Means	15
Actions via Tor Network	177
Content Unrelated to the Knowledge Structure	52
Content Related to the Knowledge Structure**	125
Content with 2 Actors per Article	14
Content with 1 Actor per Article	111
Repeated	5
Total	$111 + (14 + 14) = 139$ 139

Source: Author.

*Situations extracted from newspaper articles composed of a single actor.

**Includes articles composed of actors who are part of the knowledge structure.

²³The search, which was definitively used in the research, was conducted in the second half of February 2018. The first search was conducted in May 2017.

²⁴Therefore, the focus of our research was on the number of occurrences — not the number of articles. For detailed information, see “Appendix 8 — Articles and Occurrences for Analysis by Year.”

Defined by Strange (1988), the knowledge structure encompasses belief, established knowledge, and the communication channels through which beliefs, knowledge, and information are conveyed. These are the three aspects of the knowledge structure. This structure also determines what knowledge is discovered, how it is stored, and who communicates this knowledge—using which means, to whom, and under what conditions (STRANGE, 1988).

The anonymous TOR network is a communication channel through which beliefs, knowledge, and information are securely transmitted. Moreover, the actors originating from the 125 occurrences are embedded within the knowledge structure because they engage in actions related to the discovery, storage, and communication of information and knowledge through the TOR network. According to Strange (1988), actors positioned within the knowledge structure occupy decision-making roles. On this, she observes:

[...] Power and authority are conferred [...] on those who are acknowledged by society to be possessed of the 'right', desirable knowledge and engaged in acquisition of more of it, and on those entrusted with its storage, and on those controlling in any way the channels by which knowledge, or information, is communicated. [...] More than other structures, the power derived from the knowledge structure comes less from coercive power and more from consent, authority being conferred voluntarily on the basis of shared belief systems and the acknowledgment of the importance to the individual and to society of the particular form taken by the knowledge — and therefore of the importance of the person having the knowledge and access or control over the means by which it is stored and communicated. (STRANGE, 1988, p.121).

Actors within the knowledge structure have the capacity to influence outcomes through structural power, which, within this structure, operates less coercively and more through consent. Using this framework—journalistic articles related to the anonymous TOR network and actors positioned within the knowledge structure—we selected occurrences to construct the database. It is from this database that we analyze the diffusion of power across three dimensions: “authority,” “control,” and “outcomes.”

The column “COUNTRY” displays the name of the country where the newspaper is based; “NEWSPAPER,” the name of the online newspaper; “ARTICLE TITLE,” the title of the examined article; “AUTHOR,” the name of the author (or editor) of the newspaper article; “YEAR,” the month and year of publication in journalistic media; “ACTOR,” the actor or subject derived from the newspaper article and through which variables are examined; “CATEGORY,” the thematic group in which the subject is categorized; “PEERS,” the actor’s peers and stakeholders; “AUTHORITY,” the degree of authority of the subject in the given newspaper article;²⁵ “OBJECT,” the object controlled by the

²⁵The degrees of authority are: increasing, stable, and decreasing. The corresponding values are:

actor according to the article; “CONTROL,” the degree of control the subject has over the object in the article;²⁶ “STATUS-QUO,” the status quo before the actor’s intervention; “EFFECT ON REALITY,” whether the actor’s actions in the article had real-world effects; and “OUTCOMES,” the degree of change in the status quo in the real world caused by the subject in the given newspaper article.²⁷

Table 1.5: **Database Columns for This Research**

Column	Information
COUNTRY	Country where the newspaper is located.
NEWSPAPER	Name of the newspaper responsible for the article.
ARTICLE TITLE	Title of the article.
AUTHOR	Author of the article.
YEAR	Year of article publication.
ACTOR	Actor whose actions are described in the article.
CATEGORY	Thematic group of the occurrence.
PEERS	Identification of peers and interested parties.
AUTHORITY	Identification of the "AUTHORITY" variable in the occurrence. Receives values: "-1" for decreasing authority, "0" for stable authority, and "+1" for increasing authority.
OBJECT	Control object of the actor identified in the occurrence.
CONTROL	Identification of the "CONTROL" variable in the occurrence. Receives values: "0" for no control, "1" for partial control, and "2" for absolute control.
STATUS-QUO	Presents the status-quo of the occurrence.
EFFECT ON REALITY	Identifies whether the actor’s actions have an effect on reality. If positive, assigns "YES". If negative, assigns "NO".
OUTCOMES	Identification of the "OUTCOMES" variable in the occurrence. Receives values "0" for no change in status-quo and "1" for change in status-quo.
REPEATED ARTICLES	"YES" to indicate that the article from which the occurrence was extracted is present in another row of the database, and "NO" to indicate the article’s originality.

Source: Author.

The "CATEGORY" column was established by us and formulated to understand actors in groups of similar themes according to the journalistic articles. There are six categories:

- **Digital Security:** involves themes related to digital security.

“+1”, “0”, “-1”

²⁶The degree of control can vary between “none,” “partial,” and “total,” with the corresponding values being “0,” “1,” and “2.”

²⁷The degree of outcomes varies between “change” and “no change” in the status quo. The corresponding values are “+1” and “0.”

- **Surveillance Circumvention:** refers to themes related to shielding against digital surveillance.
- **Journalism & Whistleblowing:** relates to themes of journalism profession and whistleblowing.
- **Online Marketplace:** digital markets operating on the anonymous network.
- **Privacy:** involves news specifically about user data.

In total, we identified 25 non-state actors, which we categorized into the six aforementioned groups:

Table 1.6: Non-state Actors by Category

Category	Actors
Digital Security	Dan Egerstad, Iranian Cyber Army, Freedom Hosting, Onion Ransomware, SimpleLocker Android Malware, Ransomware
Surveillance Circumvention	Tor Project, Facebook
Journalism & Whistleblowing	Chelsea Manning, Edward Snowden, Harold T. Martin, WikiLeaks, X-Net Group, ProPublica, Strongbox, SecureDrop System
Online Marketplace	Silk Road, Silk Road 2.0, Silk Road 3.0, Alphabay, Evolution, Sheep Marketplace, Farmer's Market
Privacy	Doxbin

Source: Author.

From the newspaper articles, we extracted the necessary information to populate the database. Besides creating the categories composed of actors with similar themes, we also identified who were the peers and stakeholders interested in each actor's actions and activities. The "PEERS" column aims to identify the subjects whose degree of trust in relation to the actor increases, decreases, or remains stable. As we've seen, this trust degree is translated by the "authority" variable - hence the creation of the "PEERS" column, to provide basis for the "authority" variable. Additionally, we identified the relevant objects for achieving the intended results through the "OBJECT" column. The degree of control over the object exposed in this column translates into the "control" variable. Therefore, it was necessary to identify the control object.

The "STATUS-QUO" column seeks to identify the state of the situation prior to the subject's actions because it is from the change or maintenance of this state that we assign values to the "outcomes" variable. However, before assigning this value, we need to know whether the change or maintenance of the status-quo affects geographical reality, since

this research operates within cyberspace. If there is an effect on reality, we assign the value consistent with the change or maintenance of the status-quo. If there is no effect on reality, we assign the value "zero". Again, the filling of these fields is done based on the content of the newspaper articles.

In summary, the methodology used in this research employs both qualitative and quantitative approaches to analyze the specific objectives. The qualitative approach consists of examination of literature review; preparation and transcription of the database from alternative online newspaper sources; pre-analysis with floating readings; categorization; presentation of results.

The quantitative approach, on the other hand, focused on assigning numerical values to assist in interpreting the power diffusion in each dimension across the selected set of journalistic articles. The general methodological components were: inclusion of online newspapers for database structure; delimitation of the knowledge structure - excluding analysis of other structures presented by Strange (1988); definition of variables and their possible assigned values; period and form of data collection that composed the database.

For data analysis within the quantitative approach, the results are presented in tables, charts, graphs and descriptive statistics - through frequency calculations and relative percentages. Microsoft Excel 2010 was used for spreadsheets and calculations. Data analysis was performed through variable tabulation using descriptive statistics - with absolute and simple relative frequency. Using this method, we made considerations about the phenomenon of power diffusion in the context of the TOR anonymous network, a component of the Dark Web.

1.4.3 Database Composition for the Research

For comprehension purposes, we randomly selected a newspaper article to demonstrate how each information column was filled. The article is titled *Global Drug Survey findings: more people buying drugs online in the UK*, published on April 14, 2014 by *The Guardian*, based in the United Kingdom and authored by Ami Sedghi.

The article discusses the increase in online drug purchases in the UK and features the actor *Silk Road*, an online marketplace operating on the TOR anonymous network. According to the article, nearly 60% of respondents were aware of the Silk Road marketplace, while about 44% reported having visited the site. The peers and interested parties are private citizens.

Based on this article:

- We assigned the value "+1" (increasing authority) to the authority variable because more than half of respondents knew about the online marketplace, and the article stated that more people were buying drugs online in the UK. We interpreted that trust in the marketplace among its peers and stakeholders was growing.

- The control object of the Silk Road actor is the online marketplace itself through its webpage and Bitcoin payment mechanism. Therefore, for the control variable, we assigned the value "+2" (absolute control) since control over both the marketplace operations and payment mechanism is administered by a single actor - Silk Road itself. This control is neither dependent on nor shared with any other actor.
- We identified that the status-quo of the situation addressed by the mentioned newspaper article is the fixed number of people in the UK who consume drugs through online purchases. The Silk Road actor changes this status-quo by contributing to the increase in the number of people buying drugs online in the UK.
- This status-quo change has an effect on the real world because people are indeed consuming narcotics and illicit substances. For the outcomes variable, we assigned the value "1".
- Finally, the article in question is original - meaning an article with a single actor. There will be no other actor from this same article.

It is worth remembering that when assigning values to the variables, we also identified the peers and interested parties (authority); the relevant object for achieving desired outcomes (control); the status-quo, so we could examine whether there was any change (outcomes). These findings are fully available in the research database provided in the Dissertation Appendix.

1.5 Dissertation Structure

This dissertation is organized into three substantive chapters. The first chapter establishes the theoretical framework within International Relations (IR) that will inform the interpretation of results in the third chapter. The presentation progresses from broad conceptual foundations to specific analytical perspectives.

The first chapter comprises three principal sections. The opening section introduces British scholar Susan Strange, a seminal figure in IR and International Political Economy (IPE), providing biographical context and outlining her major contributions. The second section examines the nature of structural power through two classical IR lenses: the relational approach and the national elements approach. This dual analysis substantiates Strange's theoretical innovation regarding power concepts within IR—a departure from classical disciplinary approaches that constitutes the core of her scholarship. The third section presents Strange's framework of diffusion of power, which requires understanding both relational and structural power concepts as they operate within her structural analytical framework for IPE. This framework encompasses the economic and political structures governing people and services. We conclude the chapter with Nye's (2011)

considerations about power diffusion in the cyber domain from *The Future of Power*, complementing Strange's (1996) work which did not address cyberspace.

Chapter two focuses on the technological dimensions of information systems. The initial section discusses the technological foundations and historical development of the Internet. The second section analyzes the WWW architecture, distinguishing between the "Surface Web" and "Deep Web." Building on Sherman and Price's (2001) taxonomy, the third section classifies the "Deep Web" and situates the "Dark Web" within the broader Internet ecosystem. The chapter culminates with an explanation of the technical characteristics that distinguish the Tor network's low-latency architecture from other Dark Web networks. Tor's unique technical specifications enable operational scenarios impossible on conventional web platforms, making this network the foundational infrastructure for the diverse actors examined in this study.

The third chapter presents our analytical database and methodology for examining the diffusion of power through three operational dimensions: "authority," "control," and "outcomes." This tripartite framework operationalizes Strange's (1996) concept of diffusion of power. Our analysis assigns specific valuations to each variable, with final interpretations derived from value frequency distributions across actor groups. The chapter concludes by presenting key analytical findings.

The concluding synthesis integrates three core elements: (1) the theoretical framework, (2) the technological context, and (3) the database analysis employing our operationalized diffusion of power metrics. This integrated approach enables comprehensive examination of power diffusion phenomena within the cyber domain—specifically focusing on the Tor network ecosystem.

1.6 Final Considerations

Traditionally, International Relations (IR) emphasized two main approaches to power — relational power and material power.²⁸ The reflection on the international scenario through an analytical framework revealed a gap in power studies within IR discipline — since these approaches focused primarily on state-centric analyses. According to Strange (1996), this state-centric view is excessively restrictive as it does not correspond to the reality of the international system: transnational corporations and other non-state actors frequently exercise power that affects millions of people's daily lives, yet they remain ignored in academic analyses. As Strange herself noted:

“What the notion of structural power in world politics, society and economy did was to liberate the study of international political economy from the

²⁸Baldwin (2013) refers to these approaches as "relational" and "national elements of power," respectively.

so-called realist tradition in the study of international relations.” (Strange, 1996).

Furthermore, she argued that power diffusion adequately explained the declining quality of state authority while other actors’ authority was rapidly growing.

However, as Palan (1999) observes, Strange cannot be confined to a standard theory-practice dichotomy for two reasons: she was neither theoretical nor “empirical” in the conventional sense. Rather than being theory-oriented, Strange aimed to develop an analytical framework for exploring issues. This explains why *States and Markets* clearly presents the dilemma between theory and “theorizing”, while lacking empirical rigor (Palan, 1999). We believe this same dilemma appears in *The Retreat of the State: The Diffusion of Power in the World Economy* (1996). Despite introducing innovative themes like the accounting profession and the Mafia in global political economy, the author provides neither methodological framework nor empirical evidence to substantiate power diffusion — at least by academic standards. This observation, however, doesn’t diminish Strange’s pioneering work in IPE. As Palan (1999) suggests, perhaps strict empirical rigor would have raised her concerns about detachment from reality. She explicitly invited researchers to develop theoretical and empirical work on non-state authorities and their impact on IPE’s four structures:

“Like plants in nature, theories and explanations grow out of the dirt of observations of reality. The observations may not be ‘scientific’ in the sense that an experiment in chemistry can be objective. But they are not invented either. Getting your hands dirty with the nitty-gritty details of a technology, or with the decision-making processes of corporate strategies, or of ministerial policymaking, is a good way to test the abstractions of theory, and perhaps to develop alternate theory, or modifications of theories. Moreover, if you can illustrate a theory or a hypothesis with reference to a concrete situation, it often serves to explain more clearly the thrust of the ideas. That is part of the point of my rather scrappy descriptions of non-state authorities and how they affect power structures to be found in part II of the book. They may have been chosen somewhat at random, out of personal interest. But they are supposed to illustrate the theoretical propositions laid out in the earlier chapters. It is my sincere hope that these examples will serve to stimulate younger scholars to more innovative work, theoretical and empirical, on non-state authority in the international political economy. They are by way of being a signpost, pointing not along an open well-trodden track but rather into a mysterious forest of the unknown. Just where the path will lead, I am not at all sure. That is the nature of exploration — and its appeal to the mentally adventurous.” (STRANGE, 1996, p.xvi).

Responding to this invitation, this thesis explores studies on knowledge structure to examine non-state authorities and the diffusion of power in cyberspace, specifically within the TOR anonymous network. We pursue this through two approaches: first, by explicating three dimensions of structural power that, while not formally treated as such in Strange's works, are implicit in her writings; second, by operationalizing the diffusion of power concept through these three dimensions to enable empirical analysis. Our objective is to contribute academically to IPE studies.